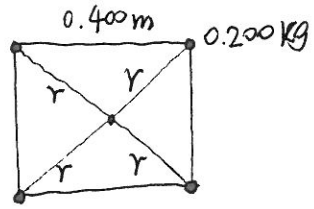
HW 09 Solutions

9.20 (a) $V = 0.250\ m/s$ so $\omega = \frac{V}{r} = \frac{0.250\ m/s}{1.25\ m} = 0.200\ rad/s = 1.91\ rpm$

(b) $a = \frac{1}{8}g = 1.225\ m/s^2$, $\alpha = \frac{a}{r} = \frac{1.225\ m/s^2}{1.25\ m} = 0.980\ rad/s^2$

(c) $s = 3.25\ m$. $\theta = \frac{s}{r} = \frac{3.25\ m}{1.25\ m} = 2.60\ rad = 149°$

9.34 (a)

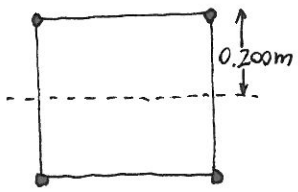


$r = \sqrt{(0.200\ m)^2 + (0.200\ m)^2} = 0.2828\ m$

$I = \sum m_i r_i^2 = 4(0.200\ kg)(0.2828\ m)^2$

$I = 0.0640\ kg \cdot m^2$

(b)

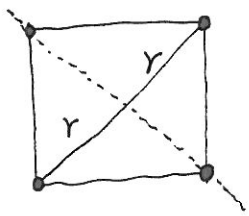


$r = 0.200\ m$

$I = \sum m_i r_i^2 = 4(0.200\ kg)(0.200\ m)^2$

$I = 0.0320\ kg \cdot m^2$

(c)



$r = 0.2828\ m$

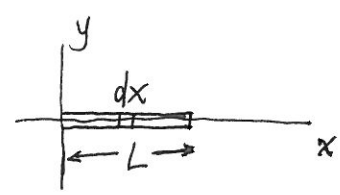
$I = \sum m_i r_i^2 = 2(0.200\ kg)(0.2828\ m)^2$

$I = 0.0320\ kg \cdot m^2$

9.36 (a) $I = 2m\left(\frac{L}{2}\right)^2 = \frac{1}{2}mL^2$

(b) $I = 2m\left(\frac{L}{4}\right)^2 + m\left(\frac{3}{4}L\right)^2 = \frac{11}{16}mL^2$

9.62 $I = \int_0^L x^2 \left(\frac{M}{L}\right) dx = \frac{M}{L}\int_0^L x^2\, dx = \frac{1}{3}ML^2$



9.67 (a) $A = \frac{a}{t} = \frac{1.80\ m/s^2}{3.00\ s} = 0.600\ m/s^3$

(b) $\alpha = \frac{a}{R} = \frac{(0.600\ m/s^3)\,t}{0.250\ m} = (2.40\ rad/s^3)\,t$

(c) $\omega_z = \int_0^t (2.40\ rad/s^3)\,t\,dt = (1.20\ rad/s^3)\,t^2$

$\omega_z = 15.0\ rad/s$ for $t = \sqrt{\frac{15.0\ rad/s}{1.20\ rad/s^3}} = 3.54\ s$

(d) $\theta - \theta_0 = \int_0^t \omega_z\, dt = \int_0^t (1.20\ rad/s^3)\,t^2\, dt = (0.400\ rad/s^3)\,t^3$

For $t = 3.54\ s$, $\theta - \theta_0 = 17.7\ rad$

9.70 (a) $K_1 + U_1 = K_2 + U_2$ gives $0 = K_{drum} + K_{mass} - mgh$.

$K_{drum} + K_{mass}$ are the same on both planets, so $mg_E h_E = mg_M h_M$.

$$h_M = h_E\left(\frac{g_E}{g_M}\right) = (5.00\,m)\left(\frac{9.80\,m/s^2}{3.71\,m/s^2}\right) = 13.2\ m$$

(b) $mg_M h_M = K_{drum} + K_{mass}$. $\frac{1}{2}mv^2 = mg_M h_M - K_{drum}$ and

$$v = \sqrt{2g_M h_M - \frac{2K_{drum}}{m}} = \sqrt{2(3.71\,m/s^2)(13.2\,m) - \frac{2(2500\,J)}{15.0\,kg}} = 8.04\,m/s$$

9.85 Gravity does work on block B and kinetic friction does work on block A.

Therefore $W_{other} = W_f = -\mu_k m_A g d$

$K_1 = 0$ (system is released from rest)

$U_1 = m_B g y_{B1} = m_B g d$; $U_2 = m_B g y_{B2} = 0$

$K_2 = \frac{1}{2}m_A v_2^2 + \frac{1}{2}m_B v_2^2 + \frac{1}{2}I\omega_2^2$

But v(blocks) $= R\omega$(pulley), so $\omega_2 = \frac{v_2}{R}$ and

$K_2 = \frac{1}{2}(m_A + m_B)v_2^2 + \frac{1}{2}I\left(\frac{v_2}{R}\right)^2$

Putting all this into the work-energy relation gives

$m_B g d - \mu_k m_A g d = \frac{1}{2}(m_A + m_B)v_2^2 + \frac{1}{2}I\left(\frac{v_2}{R}\right)^2$

$$v_2 = \sqrt{\frac{2gd(m_B - \mu_k m_A)}{m_A + m_B + I/R^2}}$$

9.86 The amount of gravitational potential energy which has become kinetic energy is

$K = (4.00\,kg - 2.00\,kg)(9.80\,m/s^2)(5.00\,m) = 98.0\,J$.

The kinetic energy of the system is $K = \frac{1}{2}(m_1 + m_2)v^2 + \frac{1}{2}I\left(\frac{v}{R}\right)^2$

$K = \frac{1}{2}(4.00\,kg + 2.00\,kg)v^2 + \frac{1}{2}(0.480\,kg\cdot m^2)\frac{v^2}{(0.160\,m)^2} = (12.4\,kg)v^2$

Solving for v gives $v = \sqrt{\frac{98.0\,J}{12.4\,kg}} = 2.81\,m/s$